

Crime Data Analysis Report

This report summarizes the initial exploratory data analysis performed on the crime data from 2020 to present.

1. Dataset Overview:

- The dataset contains over 1 million entries and 28 columns, covering details such as crime type, date and time of occurrence, location, victim demographics, and crime status.
- Key columns like `Date Rptd`, `DATE OCC` were converted to datetime objects, and `TIME OCC` was used to extract `Hour`.
- Missing values were handled by dropping rows where `Hour` could not be parsed.

2. Crime Type Distribution:

- The distribution of crime types indicates which crimes are most prevalent in the dataset. (Refer to the 'Crime Type Distribution' bar plot for specific details).

3. Yearly Crime Trend:

- The yearly crime trend shows fluctuations in the number of crimes reported over time. (Refer to the 'Yearly Crime Trend' line plot for specific details).
- There was a significant number of crimes in 2020, 2021, 2022, 2023, and a partial count for 2024 and 2025 (likely incomplete data for current years).

4. Monthly Crime Pattern:

- The monthly crime patterns reveal which months tend to have higher or lower crime rates. (Refer to the 'Monthly Crime Pattern' bar plot for specific details).

5. Crime Status Distribution:

- The pie chart for 'Crime Status Distribution' indicates the proportion of different crime statuses (e.g., 'Invest Cont' for investigation continued, 'Adj By Arrest' for adjusted by arrest).

6. Geographic Distribution (Heatmap):

- A heatmap visualizing crime locations was generated, showing areas with higher crime density. This allows for visual identification of crime hotspots.

7. High-Risk Locations:

- The top 10 locations with the highest number of reported crimes are:

- 800 N ALAMEDA ST: 2598 crimes
- 700 S FIGUEROA ST: 1711 crimes
- 100 THE GROVE DR: 1688 crimes
- 10200 SANTA MONICA BL: 1657 crimes
- 6TH ST: 1585 crimes
- 11800 SANTA MONICA BL: 1526 crimes
- 7TH ST: 1502 crimes
- 9300 TAMPA AV: 1452 crimes
- 7TH: 1447 crimes
- 6TH: 1372 crimes

This report provides a foundational understanding of the crime data, highlighting temporal trends, common crime types, and geographical hotspots.